Direct Self-evolving Optimization: Efficient Self-Evolving LLM Training without Challenger Model

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Abstract

The self-evolving training paradigm has emerged as a crucial technique for enhancing the ability of Large Language Models (LLMs) to perform mathematical reasoning and handle complex tasks. By training a challenger model to generate questions for a solver model, this approach effectively overcomes the scarcity of human-annotated data. However, a major drawback of this paradigm is the significant increase in computational and memory costs. In this paper, we eliminate the need for challenger model training by deriving a closed-form optimal challenger policy, conditioned on the current solver model. We show that this optimal policy is a reweighted version of the base model's policy, where the weights correspond to the challenger's reward. Finally, we employ the Metropolis-Hastings algorithm to iteratively sample questions from this optimal policy, which are subsequently fed into the solver model for efficient training.

1 Introduction

1.1 Existing method

In the existing self-evolving LLM training [Huang et al., 2025, Xia et al., 2025], the **challenger** optimizes:

$$\max_{\phi} R_c(\phi, \theta) := \mathbb{E}_{x \sim \pi_\phi}[r_c(x, \theta)] - \beta \mathrm{KL}(\pi_\phi || \pi_{ref}) \tag{1}$$

where $r_c(x, \theta)$ is the challenger reward function given a solver model, e.g., if a generated question x makes solver's answers more uncertain, challenger will get higher reward.

The **solver** optimizes:

$$\max_{\theta} R_s(\phi, \theta) := \mathbb{E}_{x \sim \pi_\phi}[\mathbb{E}_{y \sim \pi_\theta(y|x)}[r_s(x, y)]] \tag{2}$$

where $r_s(x, y)$ is the solver's reward function.

Since the two objectives depend on both ϕ and θ, existing solution is to fix one and update another. Given initialization ϕ_0 and θ_0, the updating rule is

$$\phi_{t+1} \leftarrow \text{update } \phi_t \text{ for K steps using GRPO on reward } R_c(\phi_t, \theta_t),$$
$$\theta_{t+1} \leftarrow \text{update } \theta_t \text{ for K steps using GRPO on reward } R_s(\phi_{t+1}, \theta_t)$$

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1.2 The Weakness and Our proposal

However, training two models are costly and this alternating optimization is unstable. To optimize equation 1 and equation 2 is to first define

$$\phi^*(\theta) := \arg\max_{\phi} R_c(\phi, \theta),$$

Indeed the closed form solution is

$$\pi_{\phi^*(\theta)} = \frac{1}{Z(\theta)} \pi_{ref}(x) \exp \frac{r_c(x, \theta)}{\beta}$$

then plug $\pi_{\phi^*(\theta)}$ into equation 2:

$$\begin{aligned}
\max_{\theta} R(\theta) &:= R_s(\phi^*(\theta), \theta) \\
&= \mathbb{E}_{x \sim \pi_\phi}[\mathbb{E}_{y \sim \pi_\theta}[r_s(x, y)]] = \sum_x \pi_{\phi^*(\theta)}(x|p_0) \mathbb{E}_{x \sim \pi_\theta}[r_s(x, y)]] \\
&= \sum_x \frac{1}{Z(\theta)} \pi_{ref}(x|p_0) \exp \frac{r_c(x, \theta)}{\beta} \mathbb{E}_{y \sim \pi_\theta(y|x)}[r_s(x, y)] \\
&= \frac{1}{Z(\theta)} \mathbb{E}_{x \sim \pi_{ref}(x|p_0)} \left[\exp \frac{r_c(x, \theta)}{\beta} \mathbb{E}_{y \sim \pi_\theta(y|x)}[r_s(x, y)] \right]
\end{aligned} \tag{3}$$

where $r_c(x, \theta) = \max 0, r_{\text{uncertainty}}(x; \theta) - r_{\text{rep}}(x)$, $r_{\text{uncertainty}}(x; \theta) := 1 - 2|\hat{p}(x; S_\theta(x)) - \frac{1}{2}|$, and $\hat{p}(x; S_\theta(x)) = \frac{1}{m} \sum_{j=1}^m \mathbb{I} y_j = \bar{y}(x)$, and $S_\theta(x) = y_1, ... y_m$ is the set of m responses that solver model θ generated for question x, and $\bar{y}(x)$ is the most frequent answer in $S_\theta(x)$.

$r_{\text{rep}}(x)$ is the repetition penalty. Given a batch of questions $x_1, \ldots, x_B$, we compute pairwise distances using BLEU score similarity, $d_{ij} = 1 - BLEU(x_i, x_j)$, and group questions where $d_{ij} < \tau_{BLEU}$ into clusters $C = \{C_1, ..., C_K\}$. The penalty for a question x_i in a cluster C_k is proportional to its relative size:

$$r_{\text{rep}}(x) = \frac{|C_k|}{B}.$$

For $r_s(x, y_j)$, it is 1 if y_j is identical to the pseudo label $\bar{y}(x)$, else 0.

Hence, we can directly optimize this objective using policy gradient. First, we generate a mini-batch of questions from the base model π_{ref}: $\{x_1, ..., x_G\}$. For each question, the solver model generates m responses: $A = \{y_{i,1}, ..., y_{i,m}\}$. Then we estimate the policy gradient as

$$\hat{\nabla}_\theta R(\theta) = \frac{1}{G} \sum_{i=1}^{G} \exp \frac{r_c(x_i, \theta)}{\beta} \frac{1}{m} \sum_{j=1}^{m} \nabla_\theta \log \pi_\theta(y_{i,j}|x_i) \tag{4}$$

We directly optimize the solver model. It is a weighted policy gradient method, where the weights depend on how the current solver model performs on the generated questions. For the questions that the solver has higher uncertainty, its gradient will be up-weighted.

1.3 Active Sampling

The objective in equation 3 can be equivalently written as:

$$\max_{\theta} R(\theta) = \mathbb{E}_{x \sim \pi_{\phi^*(\theta)}(x|p_0)} \left[\mathbb{E}_{y \sim \pi_\theta(y|x)}[r_s(x, y)] \right]$$

where $\pi_{\phi^*(\theta)} \sim \pi_{ref}(x) \exp \frac{r_c(x,\theta)}{\beta}$. Now the question is, how can we actively sample questions from $\pi_{\phi^*(\theta)}(x|p_0)$ only given access to π_{base}? A generally used approach is MCMC sampling via Metropolis-Hasting algorithm.

Algorithm 1: MCMC-DEO

Input: θ_0, question-generating prompt p_0, learning rate η, regulariztaion parameter β.

for $t = 0, \dots, T-1$ **do**

  Sample N questions $x_1^0, \dots, x_N^0$ from $\pi_{base}(\cdot|p_0)$.

  For each x_i^0, generate mutated version $\hat{x}_i = MCMC(x_i^0, \tau, \pi_{base})$ For each $\hat{x}_i$, generate m answers $y_{i,1}, \dots, y_{i,m}$ from $\pi_{\theta_t}(\cdot|x_i)$.

  Compute mini-batch gradient $\hat{\nabla}_\theta R(\theta) = \frac{1}{N}\sum_{i=1}^N \frac{1}{m}\sum_{j=1}^m \nabla_\theta r_s(x_i, y_{i,j}) \log \pi_\theta(y_{i,j}|x_i)$.

  $\theta_{t+1} = \theta_t + \eta\hat{\nabla}_\theta R(\theta)$.

end

Output: θ_T

Algorithm 2: MCMC(x_0, τ, π)

Input: Original question x_0, mutation prompt p, policy π

for $t = 0, \dots, \tau - 1$ **do**

  Sample a mutated question $x_{t+1} \sim \pi(x_t|p)$,

  Compute acceptance probability $\alpha \doteq \min 1, \exp(r_c(x') - r_c(x))$

  Accepting $x_{t+1} = x'$ with probability α, otherwise $x_{t+1} = x_t$.

end

Output: x_τ

References

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Peng Xia, Kaide Zeng, Jiaqi Liu, Can Qin, Fang Wu, Yiyang Zhou, Caiming Xiong, and Huaxiu Yao. Agent0: Unleashing self-evolving agents from zero data via tool-integrated reasoning. *arXiv preprint arXiv:2511.16043*, 2025.

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